



COEP Technological University

(COEP Tech)

A Unitary Public University of Government of Maharashtra

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(Formerly College of Engineering Pune)

YEAR : SECOND YEAR

SEMISTER: III

SUBJECT: Geo-informatics

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Attribute Data & Tables

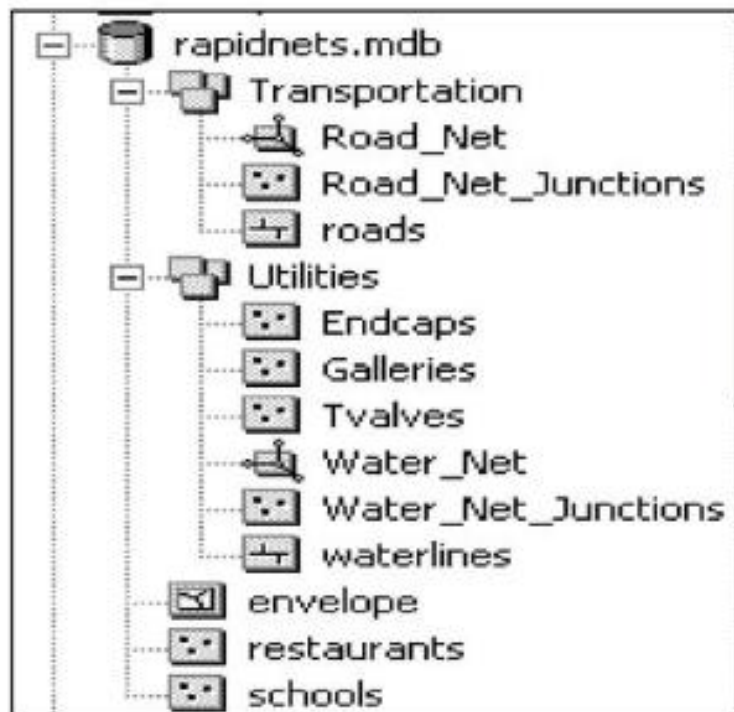
What is network?

- A network is a system of interconnected elements, such as edges (lines) and connecting junctions (points), that represent possible routes from one location to another.
- People, resources, and goods tend to travel along networks: cars and trucks travel on roads, airliners fly on predetermined flight paths, oil flows in pipelines. By modeling potential travel paths with a network, it is possible to perform analyses related to the movement of the oil, trucks, or other agents on the network.
- The most common network analysis is finding the shortest path between two points.
- ArcGIS groups networks into two categories:
 - geometric networks, and
 - network datasets.

What is networks?

- Any system of interconnected linear features is a network.

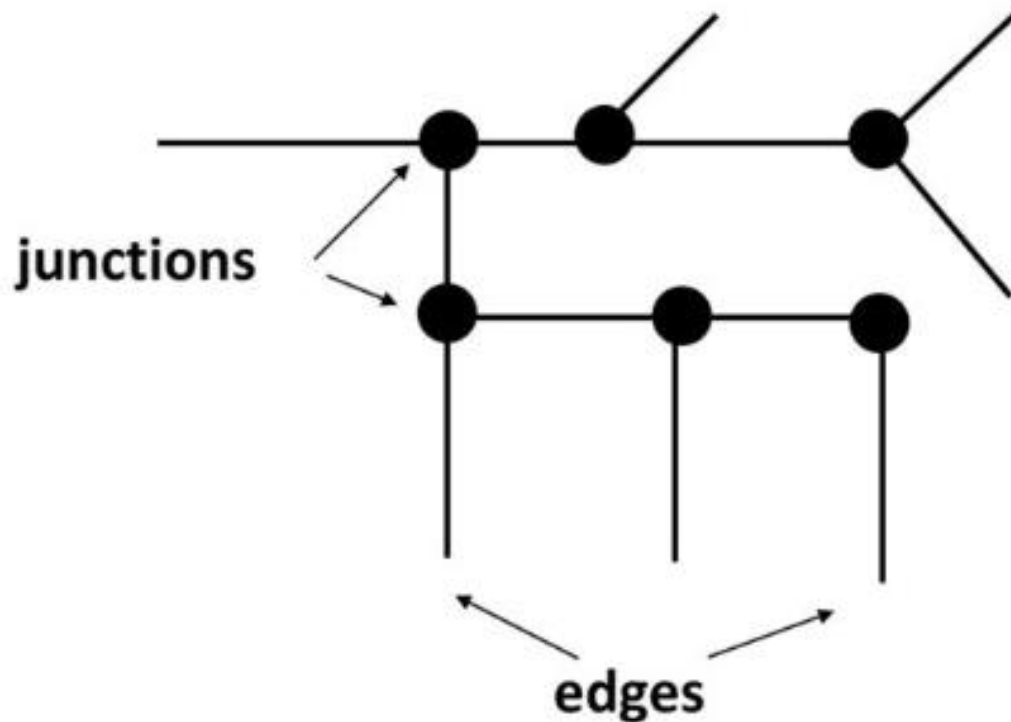
Geometric networks



- Built inside a feature dataset
- May have many participating feature classes
- Requires ArcEditor or ArcInfo to build

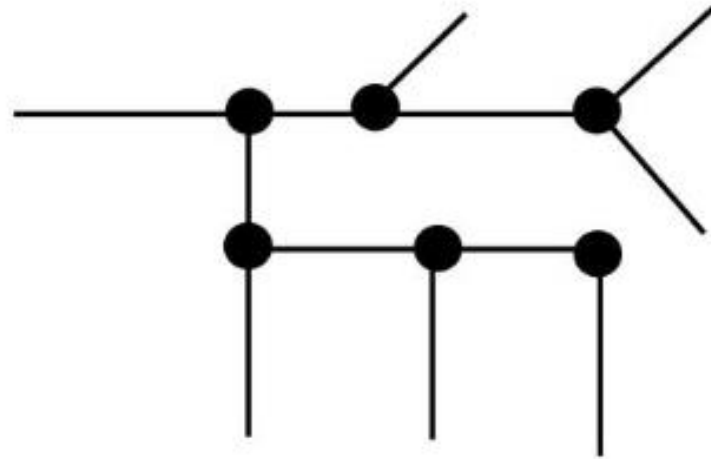
Network components

- Composed of edges and junctions



Network structure

Geometric network is composed of feature classes



Logical network consists of tables describing network relationships



Types of networks

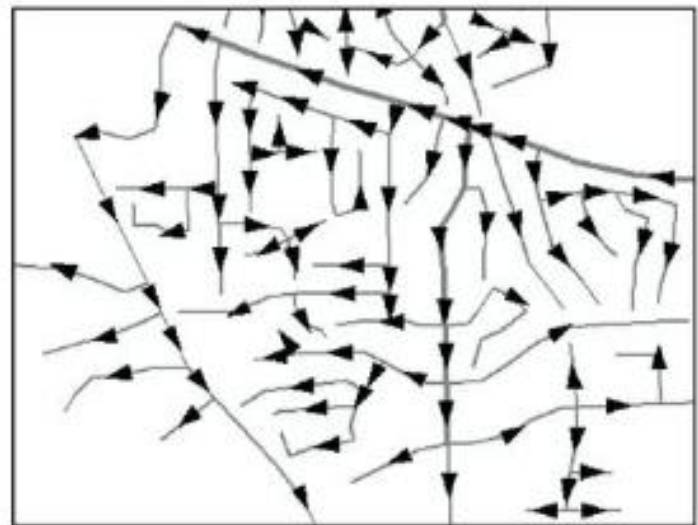
Transportation Networks

Material moves as it choose



Utility Networks

Material flow dictated by network geometry and sources/sinks



Network problems

- Transportation
 - What is the best path to travel to sixteen delivery locations?
 - What is the likely service area of a fire station based on travel time?
 - What is the shortest path from point A to point B?
 - What is the shortest path which avoids narrow streets?
- Utility networks
 - If a valve fails, which customers will be affected?
 - If I have to close this pipe for repairs, can I reroute water through another path to minimize service disruption?
 - How will contamination at one location propagate through the network?
 - Which sewer lines serve only residential customers?

Types of network analysis layers

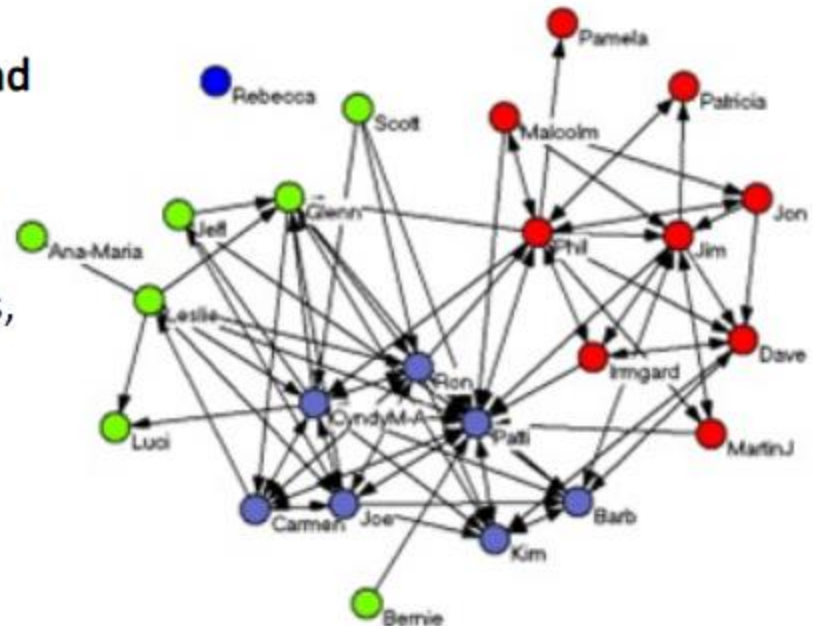
- ArcGIS Network Analyst allows you to solve common network problems, such as finding the best route across a city, finding the closest emergency vehicle or facility, identifying a service area around a location, servicing a set of orders with a fleet of vehicles, or choosing the best facilities to open or close. 6 types or solvers:
 - Route
 - Closest facility
 - Service areas
 - OD cost matrix (an origin-destination (OD) cost matrix from multiple origins to multiple destinations)
 - Vehicle routing problem
 - Location-allocation

Network

- A network is a system of linear features that has the appropriate attributes for the flow of objects.
- A network is typically topology-based: lines (arcs) meet at intersections (junctions), lines cannot have gaps, and lines have directions.
- Attribute data of a road network include link impedance, turns, one-way streets, and overpasses/underpasses.

Explain the difference between a network and a line shapefile.

Unlike a line shapefile, a network is typically topology-based: lines (arcs) meet at intersections (nodes), lines cannot have gaps, and lines have directions.



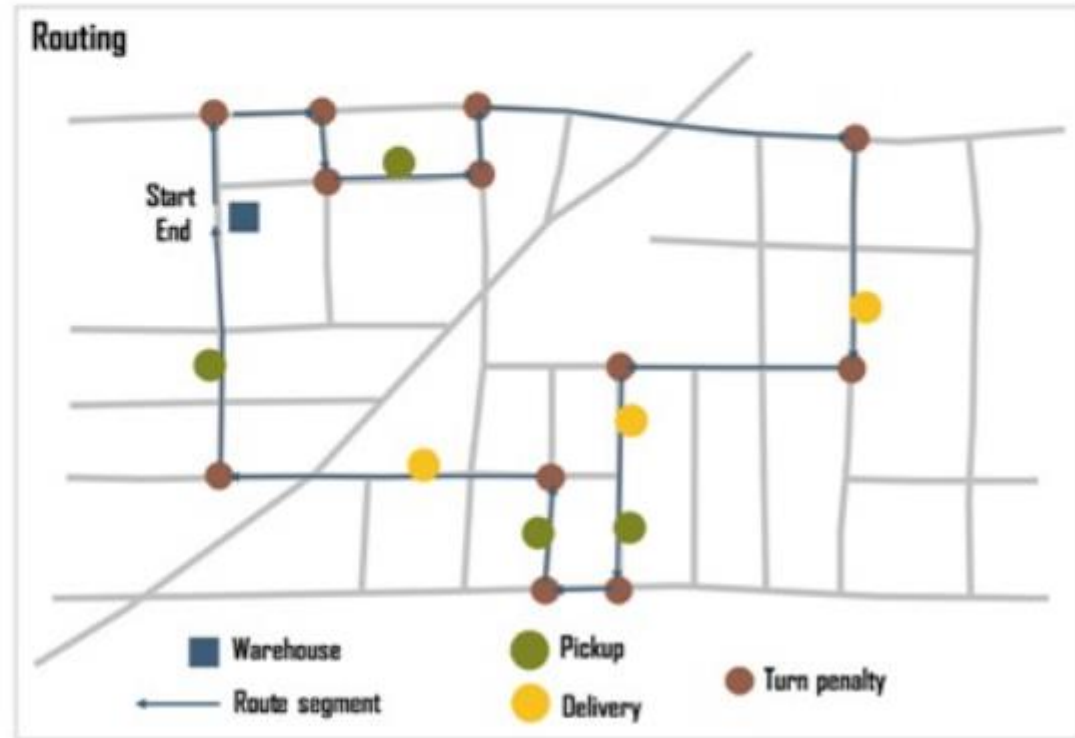
Link and Link Impedance

- A link refers to a road segment defined by two end points. Links are the basic geometric features of a network.
- Link impedance is the cost of traversing a link.

What is link impedance?

Link impedance is the cost of traversing a link.

A simple measure of the cost is the physical length of the link.



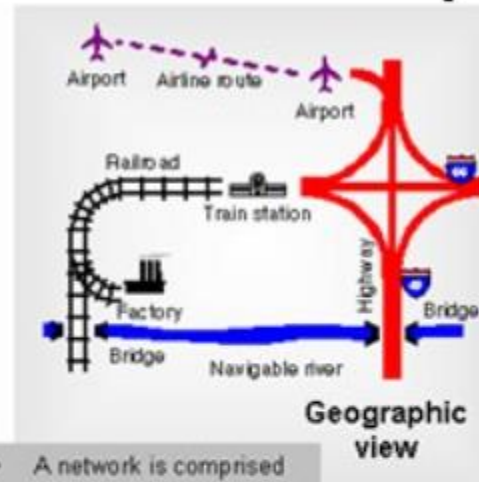
Junction and Turn Impedance

■ A junction refers to a street intersection.

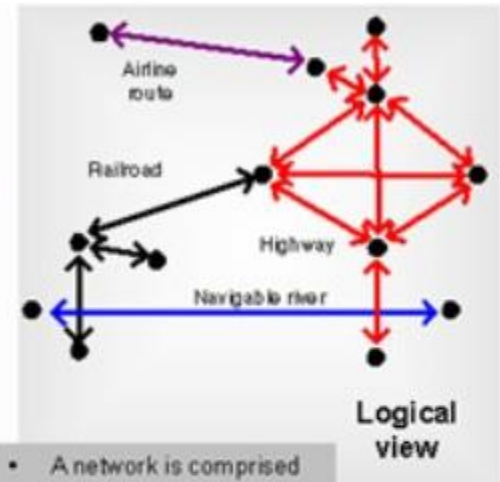
■ A turn is a transition from one street segment to another at a junction.

■ Turn impedance is the time it takes to complete a turn, which is significant in a congested street network. Turn impedance is directional.

■ A turn table assigns the turn impedance value to each turn in the network.



- A network is comprised of linear features through which commodities flow
- A network contains point features where a transfer of commodities occur, such as from roads to railroad stations, or from one road to another.



- A network is comprised of edges and junctions.
- Commodities flow through edges, and edges connect together at junctions, where flow from one edge is transferred to another.
- In a network, geometry is not important, only the connectivity of edges and junctions.

What is turn impedance?

Turn impedance is the time it takes to complete a turn, which is significant in a congested street network

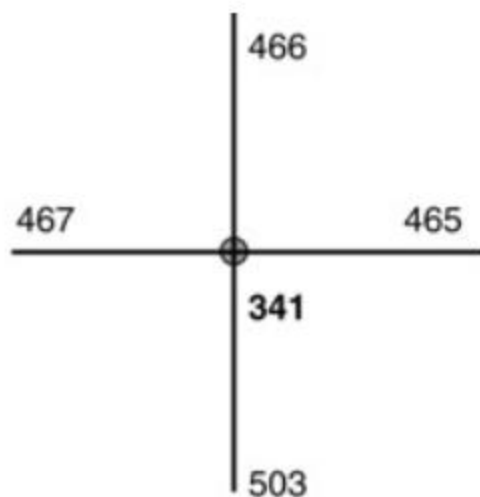


Figure 17.6
Possible turns at node 341.

Node#	Arc1#	Arc2#	Angle	Minutes
341	503	467	90	0.500
341	503	466	0	0.250
341	503	465	-90	0.250
341	467	503	-90	0.250
341	467	466	90	0.500
341	467	465	0	0.250
341	466	503	0	0.250
341	466	467	-90	0.250
341	466	465	90	0.500
341	465	503	90	0.500
341	465	467	0	0.250
341	465	466	-90	0.250

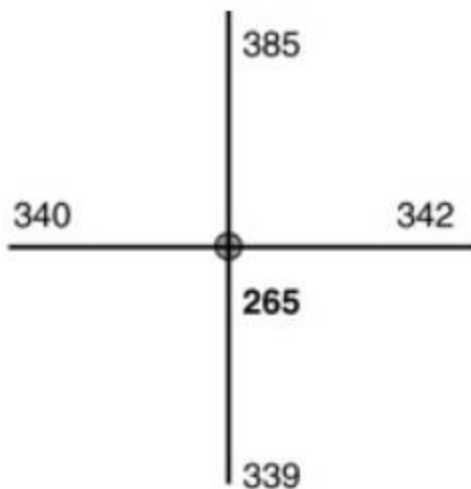


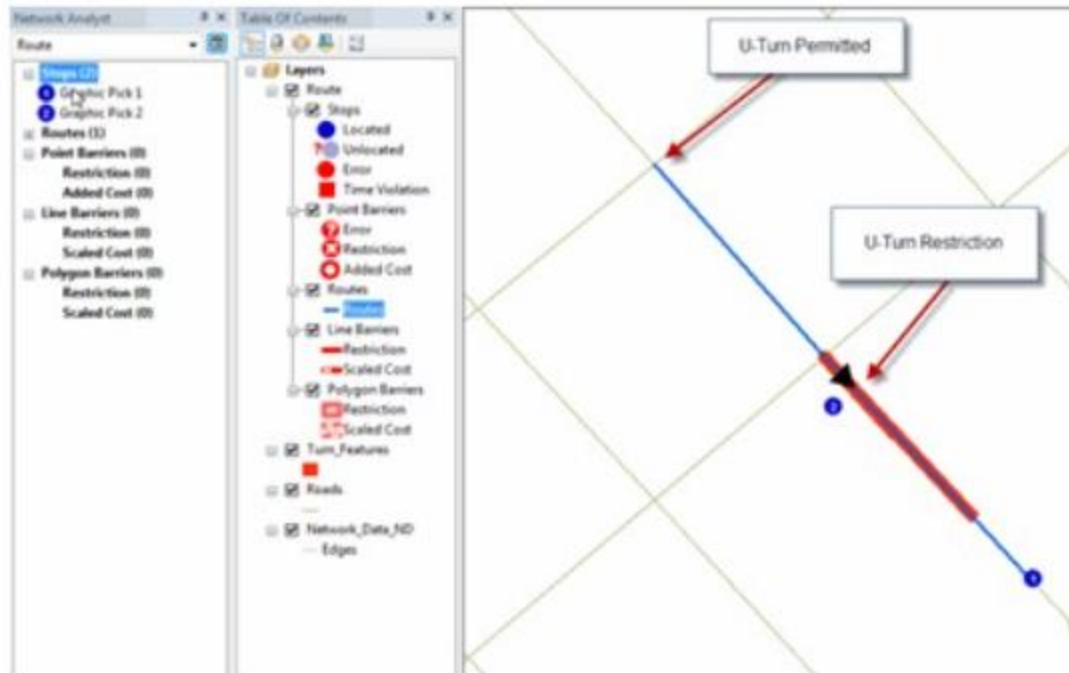
Figure 17.7

Node 265 has stop signs for the east–west traffic. Turn impedance applies only to turns in the shaded rows.

Node#	Arc1#	Arc2#	Angle	Minutes
265	339	342	-87.412	0.000
265	339	340	92.065	0.000
265	339	385	7.899	0.000
265	342	339	87.412	0.500
265	342	340	-0.523	0.250
265	342	385	-84.689	0.250
265	340	339	-92.065	0.250
265	340	342	0.523	0.250
265	340	385	95.834	0.500
265	385	339	-7.899	0.000
265	385	342	84.689	0.000
265	385	340	-95.834	0.000

Restrictions

Restrictions refer to routing requirements on a network. One-way or closed streets are examples of restrictions.

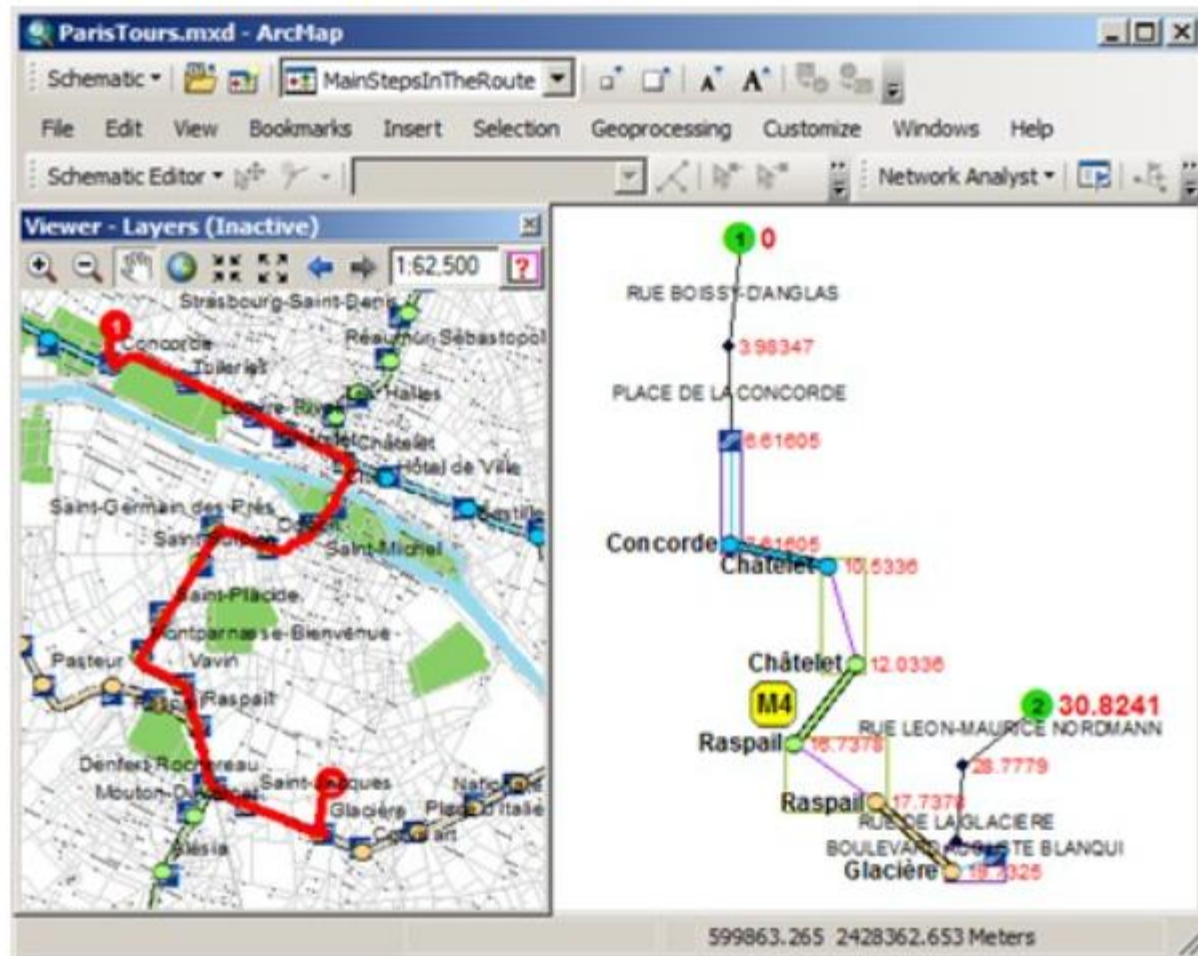


What are considered restrictions in network analysis?

One-way or closed streets are considered restrictions in network analysis

Putting Together a Network

1. Gathering linear features from a network data source
2. Editing and building network
3. Attributing the network features



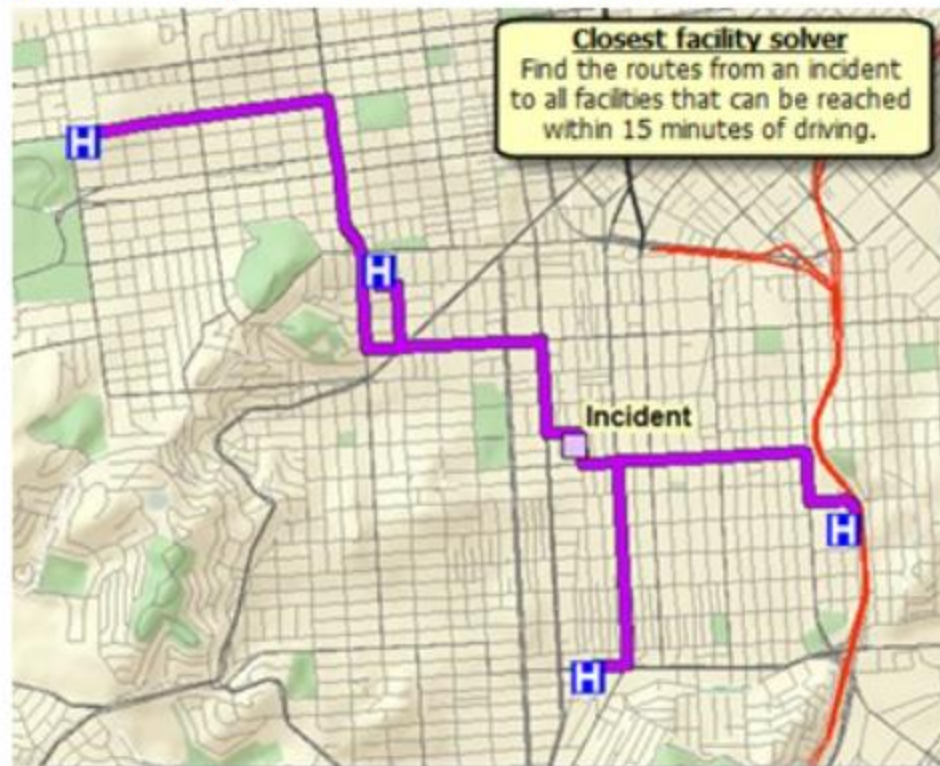
Network analysis

A network with the appropriate attributes can be a variety of applications including shortest path analysis, traveling salesman problem, vehicle routing problem, closest facility, allocation, and location-allocation.



Shortest Path Analysis

Shortest path analysis finds the path with the minimum cumulative impedance between nodes on a network. The path may connect just two nodes—an origin and a destination—or have specific stops between the nodes.



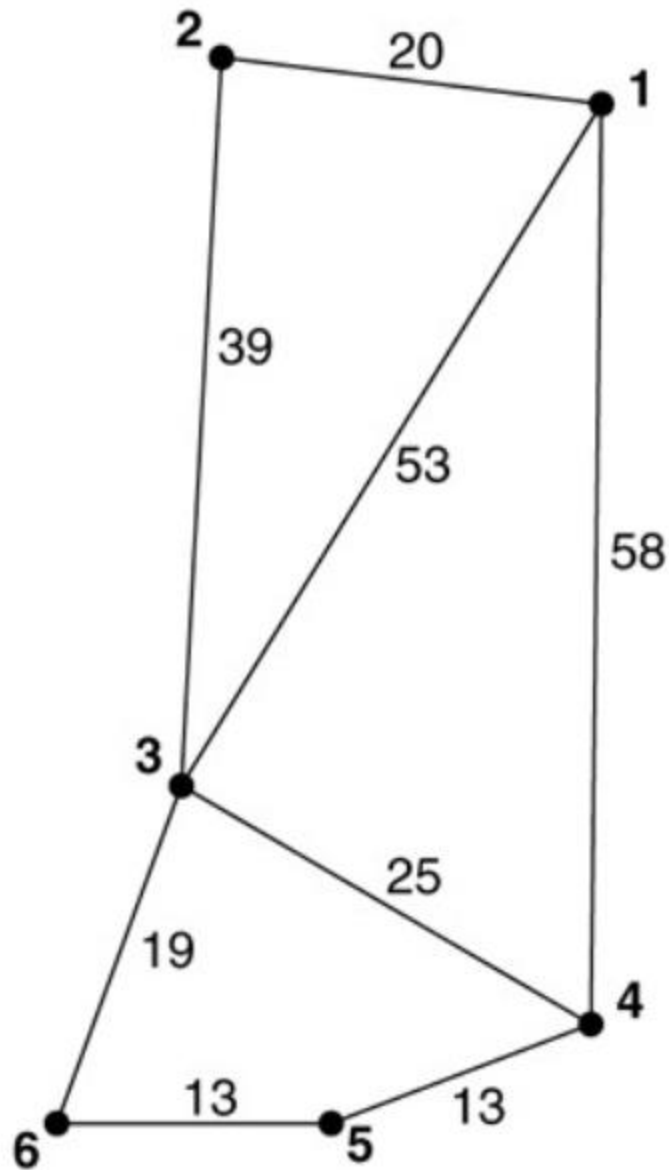


Figure 17.8
Link impedance
values between cities
on a road network.

	(1)	(2)	(3)	(4)	(5)	(6)
(1)	∞	20	53	58	∞	∞
(2)	20	∞	39	∞	∞	∞
(3)	53	39	∞	25	∞	19
(4)	58	∞	25	∞	13	∞
(5)	∞	∞	∞	13	∞	13
(6)	∞	∞	19	∞	13	∞

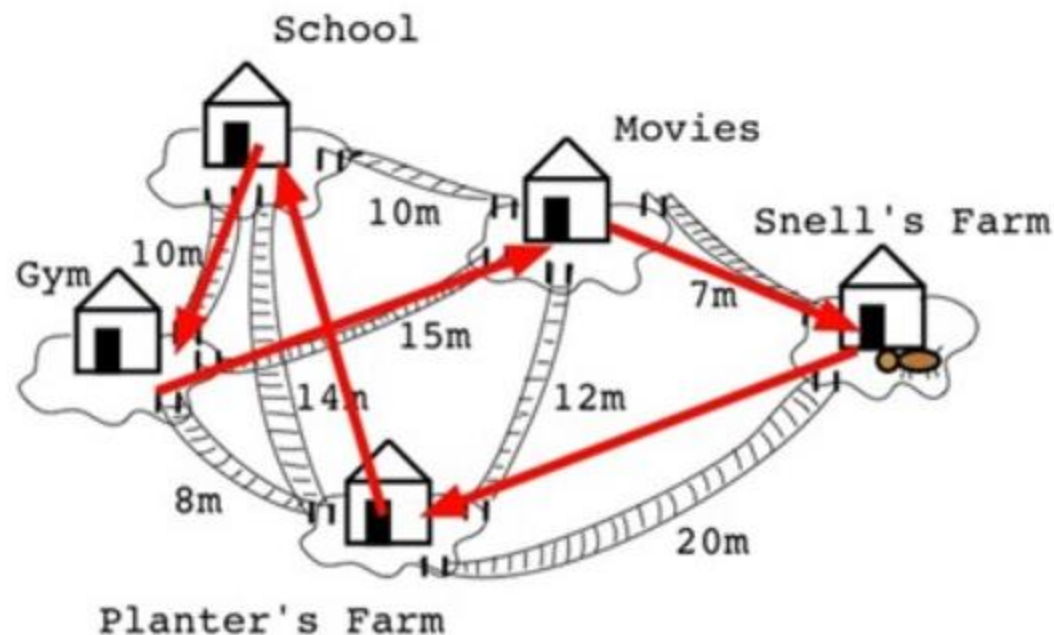
TABLE 17.1 The Impedance Matrix among Six Nodes in Figure 17.11

From-node	To-node	Shortest Path	Minimum Cumulative impedance
1	2	p_{12}	20
1	3	p_{13}	53
1	4	p_{14}	58
1	5	$p_{14} + p_{45}$	71
1	6	$p_{13} + p_{36}$	72

TABLE 17.2 Shortest Paths from Node 1 to All Other Nodes in Figure 17.11

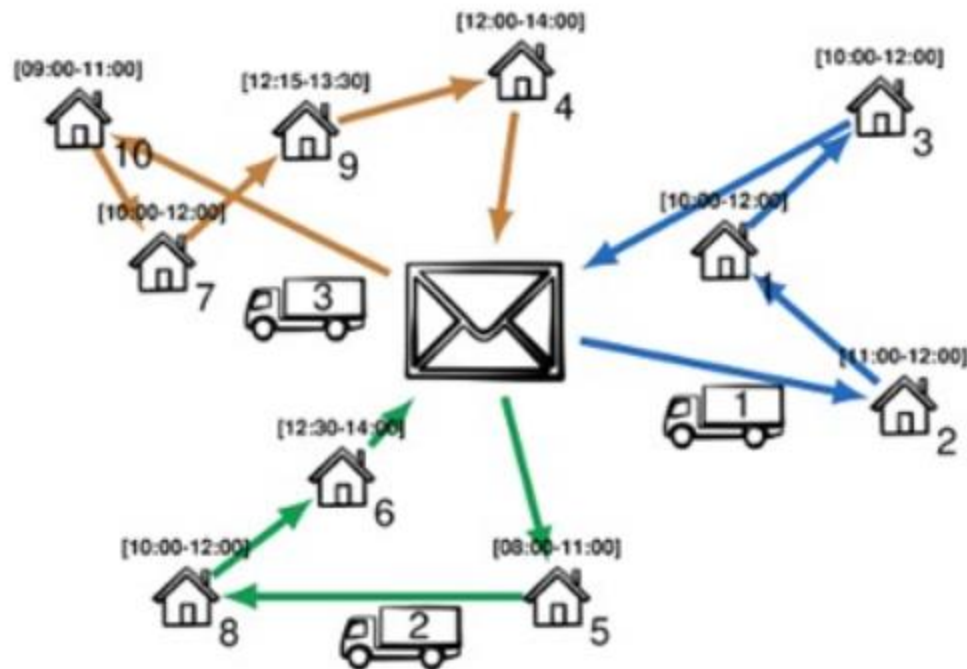
Traveling Salesman Problem

The traveling salesman problem is a routing problem that adds two constraints to the shortest path analysis: The salesman must visit each of the select stops only once, and the salesman may start from any stop but must return to the original stop.



Vehicle Routing Problem

- Given a fleet of vehicles and customers, the main objective of the vehicle routing problem is to schedule vehicle routes and visits to customers in such a way that the total travel time is minimized.
- Additional constraints such as time windows, vehicle capacity, and dynamic conditions (e.g., traffic congestion) may also exist.



Closest Facility

Closest facility finds the closest facility, such as a hospital, fire station, or ATM, to any location on a network.





Figure 17.9
Shortest path from
a street address to
its closest fire
station, shown by
the square symbol.

Allocation

Allocation measures the efficiency of public facilities, such as fire stations, or school resources, in terms of their service areas.



The result of an allocation analysis is typically presented as service areas. Why?

Allocation analysis works with the spatial distribution of resources such as fire stations and schools. Therefore, the result of an allocation analysis is typically presented as service areas, indicating the area extents of services that can be rendered by these resources.



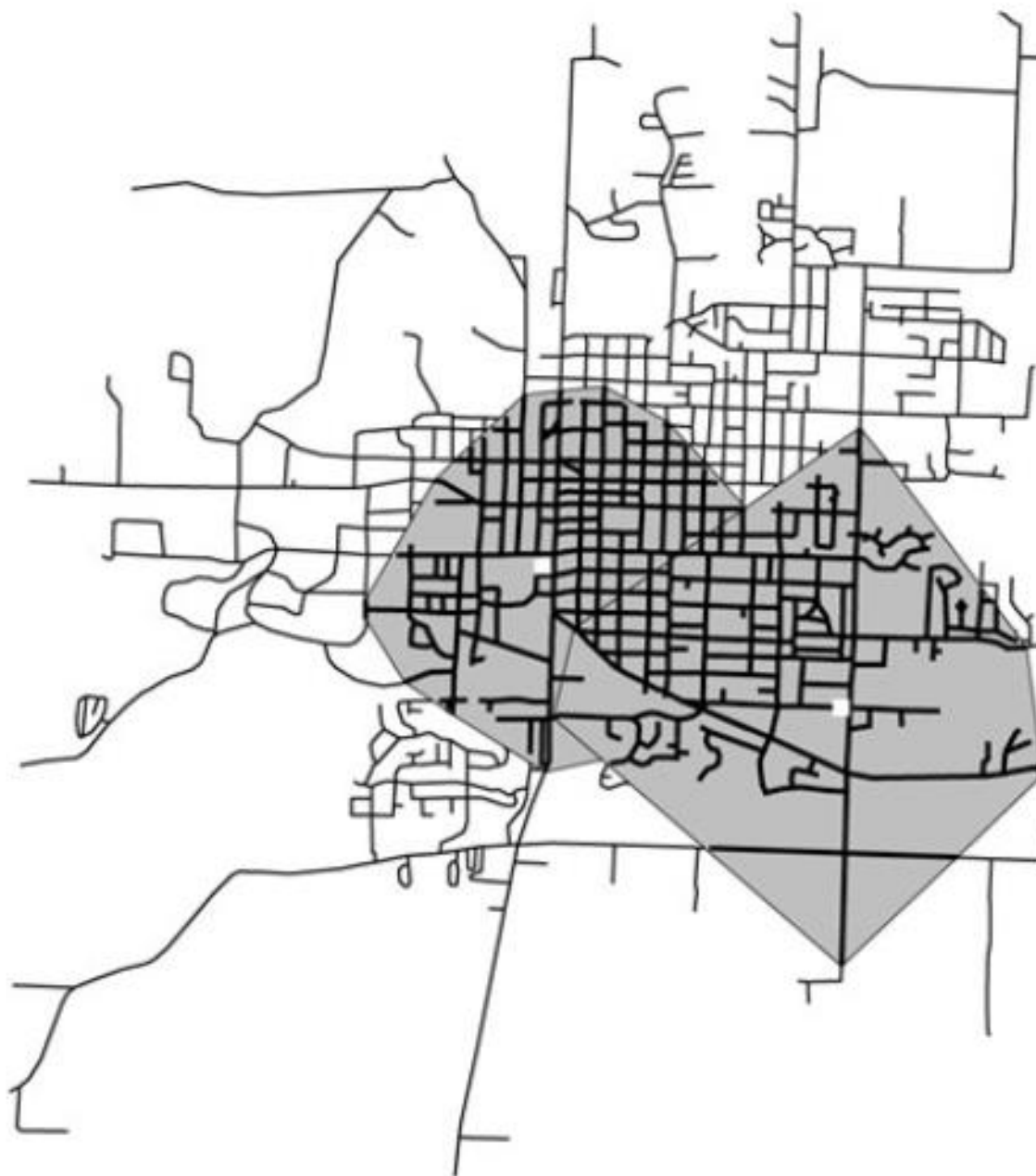


Figure 17.10
Service areas of
two fire stations
within a 2-minute
response time.

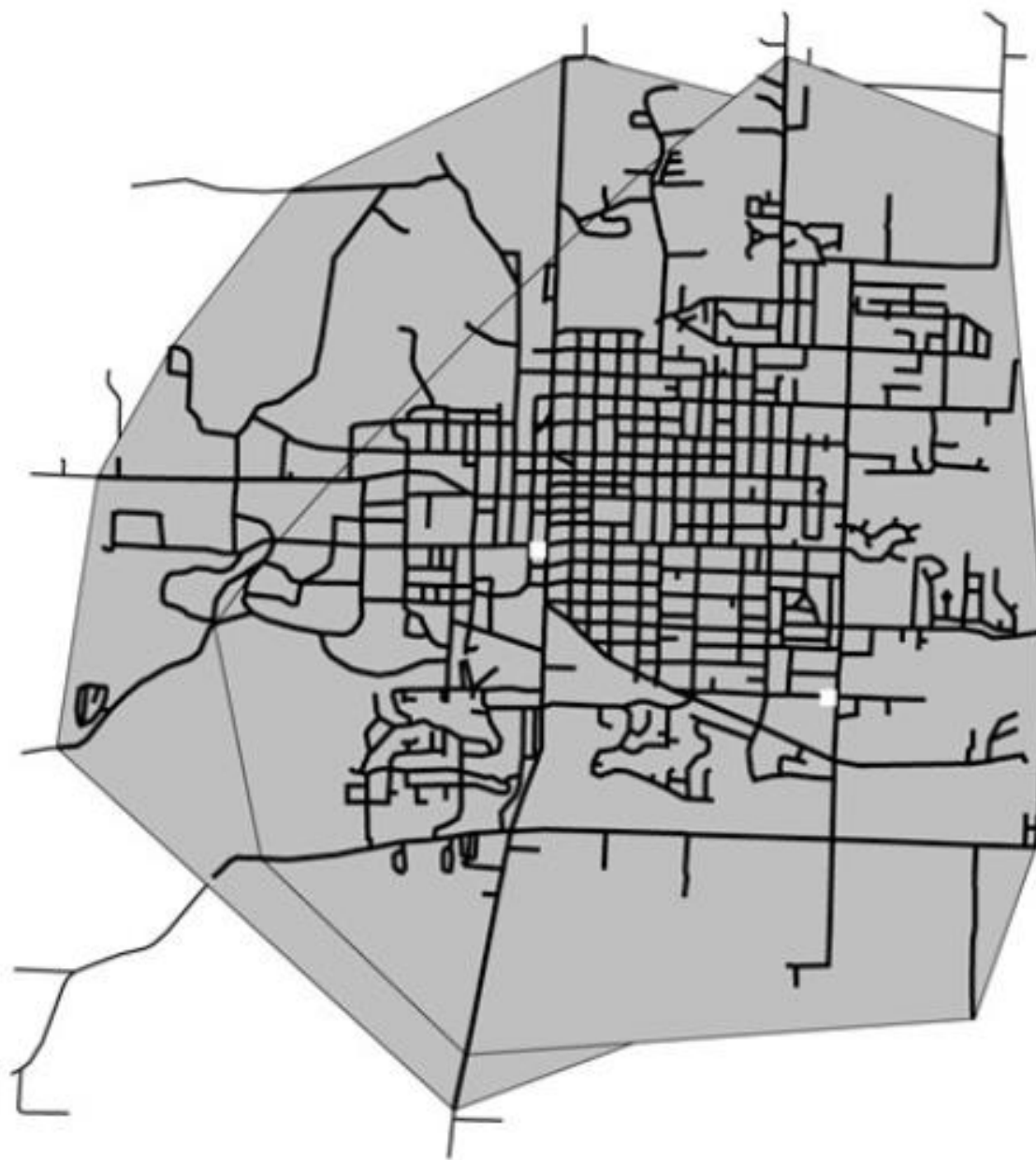


Figure 17.11
Service areas of
two fire stations
within a 5-minute
response time.

Location-Allocation

Location-allocation solves problems of matching the supply and demand by using sets of objectives and constraints.



Define location-allocation analysis. Location-allocation analysis solves problems of matching the supply and demand by using sets of objectives and constraints.



Figure 17.12

The two solid squares represent existing fire stations, the three open squares candidate facilities, and the seven circles nursing homes. The map shows the result of matching two existing fire stations with nursing homes based on the minimum impedance model and an impedance cutoff of 4 minutes on the road network.

Thank Your